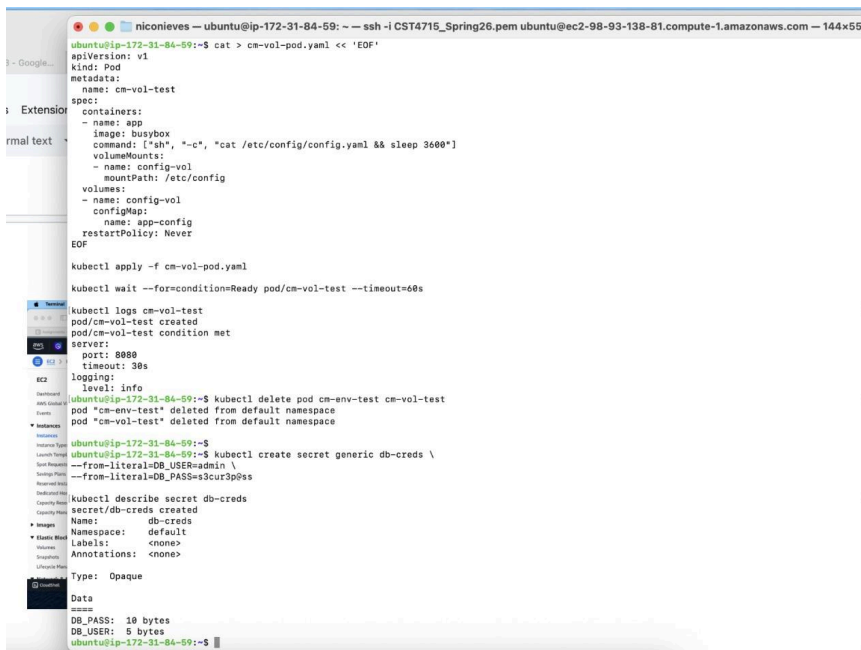
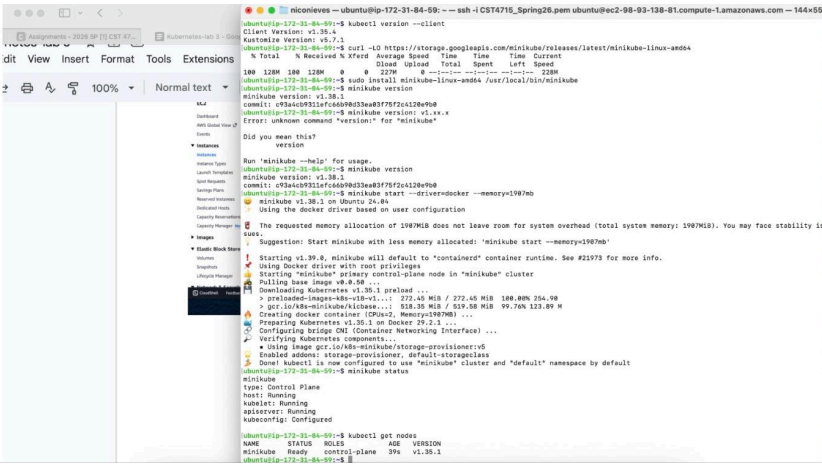
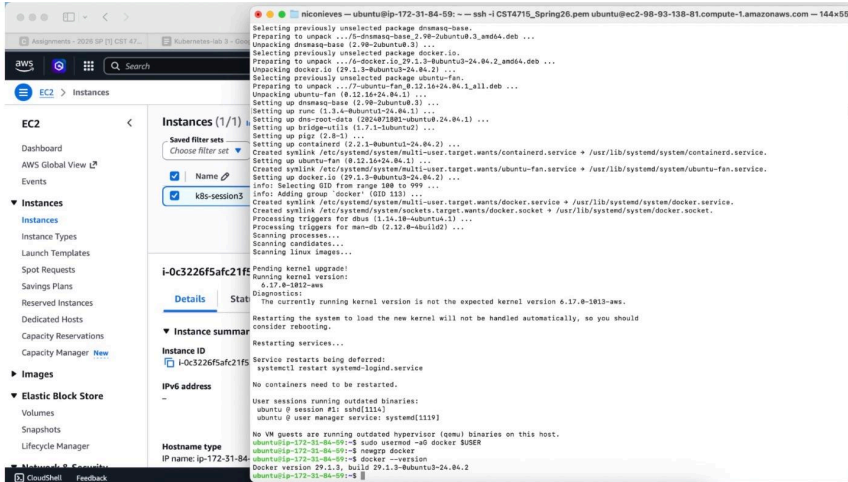




```
ubuntu@ip-172-31-84-89:~$ kubectl delete pod ce-env-test ce-vol-test
pod "ce-env-test" deleted from default namespace
pod "ce-vol-test" deleted from default namespace

ubuntu@ip-172-31-84-89:~$ kubectl create secret generic db-creds \
--from-literal=DB_USER=admin \
--from-literal=DB_PASS=skur3p05s

kubectl describe secret db-creds
secret/db-creds created
Name:         db-creds
Namespace:    default
Labels:       <none>
Annotations:  <none>

Type: Opaque

Data
====
DB_PASS: 16 bytes
DB_USER: 8 bytes
ubuntu@ip-172-31-84-89:~$ kubectl get secret db-creds -o jsonpath='{.data.DB_USER}' | base64 -d
adminubuntu@ip-172-31-84-89:~$ kubectl get secret db-creds -o jsonpath='{.data.DB_PASS}' | base64 -d
skur3p05s
ubuntu@ip-172-31-84-89:~$ cat > secret-pod.yaml << EOF
apiVersion: v1
kind: Pod
metadata:
  name: secret-test
spec:
  containers:
    - name: app
      image: busybox
      command: ["sh", "-c", "echo User: DB_USER && echo Pass: DB_PASS && sleep 3600"]
      env:
        - name: DB_USER
          valueFrom:
            secretKeyRef:
              name: db-creds
              key: DB_USER
        - name: DB_PASS
          valueFrom:
            secretKeyRef:
              name: db-creds
              key: DB_PASS
      restartPolicy: Never
EOF
ubuntu@ip-172-31-84-89:~$ kubectl apply -f secret-pod.yaml
pod/secret-test created
ubuntu@ip-172-31-84-89:~$ kubectl wait --for=condition=Ready pod/secret-test --timeout=60s
pod/secret-test condition met
ubuntu@ip-172-31-84-89:~$ kubectl logs secret-test
User: admin
Pass: skur3p05s
ubuntu@ip-172-31-84-89:~$ kubectl delete pod secret-test
kubectl delete secret db-creds
pod "secret-test" deleted from default namespace
kubectl get storageclass
secret "db-creds" deleted from default namespace
ubuntu@ip-172-31-84-89:~$ kubectl get storageclass
NAME                                PROVISIONER                RECLAIMPOLICY   VOLUMEBINDINGMODE   ALLOWVOLUMEEXPANSION   AGE
standard (default)                  k8s.io/minikube-hostpath   Delete           Immediate            false                   14m
ubuntu@ip-172-31-84-89:~$
```

```
Data
====
DB_PASS: 16 bytes
DB_USER: 8 bytes
ubuntu@ip-172-31-84-89:~$ kubectl get secret db-creds -o jsonpath='{.data.DB_USER}' | base64 -d
adminubuntu@ip-172-31-84-89:~$ kubectl get secret db-creds -o jsonpath='{.data.DB_PASS}' | base64 -d
skur3p05s
ubuntu@ip-172-31-84-89:~$ cat > secret-pod.yaml << EOF
apiVersion: v1
kind: Pod
metadata:
  name: secret-test
spec:
  containers:
    - name: app
      image: busybox
      command: ["sh", "-c", "echo User: DB_USER && echo Pass: DB_PASS && sleep 3600"]
      env:
        - name: DB_USER
          valueFrom:
            secretKeyRef:
              name: db-creds
              key: DB_USER
        - name: DB_PASS
          valueFrom:
            secretKeyRef:
              name: db-creds
              key: DB_PASS
      restartPolicy: Never
EOF
ubuntu@ip-172-31-84-89:~$ kubectl apply -f secret-pod.yaml
pod/secret-test created
ubuntu@ip-172-31-84-89:~$ kubectl wait --for=condition=Ready pod/secret-test --timeout=60s
pod/secret-test condition met
ubuntu@ip-172-31-84-89:~$ kubectl logs secret-test
User: admin
Pass: skur3p05s
ubuntu@ip-172-31-84-89:~$ kubectl delete pod secret-test
kubectl delete secret db-creds
pod "secret-test" deleted from default namespace
kubectl get storageclass
secret "db-creds" deleted from default namespace
ubuntu@ip-172-31-84-89:~$ kubectl get storageclass
NAME                                PROVISIONER                RECLAIMPOLICY   VOLUMEBINDINGMODE   ALLOWVOLUMEEXPANSION   AGE
standard (default)                  k8s.io/minikube-hostpath   Delete           Immediate            false                   14m
ubuntu@ip-172-31-84-89:~$
```

```
accessModes:
  - ReadWriteOnce
storageClassName: standard
resources:
  requests:
    storage: 2Gi
EOF

kubectl apply -f mysql-pvc.yaml
kubectl get pvc mysql-pvc
persistentvolumeclaim/mysql-pvc created
NAME      STATUS   VOLUME                                     CAPACITY   ACCESS MODES   STORAGECLASS   VOLUMEATTRIBUTESCLASS   AGE
mysql-pvc   Bound    pvc-74961da9-2ade-4cbb-b3dd-dca3f88643ba   2Gi        RWO            standard              <unset>                  8s
ubuntu@ip-172-31-84-89:~$ cat > mysql-deploy.yaml << EOF
apiVersion: apps/v1
kind: Deployment
metadata:
  name: mysql
spec:
  replicas: 1
  selector:
    matchLabels:
      app: mysql
  template:
    metadata:
      labels:
        app: mysql
    spec:
      containers:
        - name: mysql
          image: mysql:8.0
          envFrom:
            - secretRef:
                name: mysql-secret
            - configMapRef:
                name: mysql-config
          ports:
            - containerPort: 3306
          volumeMounts:
            - name: data
              mountPath: /var/lib/mysql
          volumes:
            - name: data
              persistentVolumeClaim:
                claimName: mysql-pvc
EOF
ubuntu@ip-172-31-84-89:~$ kubectl apply -f mysql-deploy.yaml
deployment.apps/mysql created
ubuntu@ip-172-31-84-89:~$ kubectl rollout status deployment/mysql
Waiting for deployment 'mysql' rollout to finish: 0 of 1 updated replicas are available...
deployment 'mysql' successfully rolled out
ubuntu@ip-172-31-84-89:~$ kubectl exec -it $(kubectl get pods -l app=mysql -o name | head -1) -- mysql -uroot -prootpass -e 'SHOW DATABASES;'
mysql: [Warning] Using a password on the command line interface can be insecure.
+-----+
| Database |
+-----+
| appdb    |
| information_schema |
| mysql    |
| performance_schema |
| sys      |
+-----+
ubuntu@ip-172-31-84-89:~$
```